

CSS Theming Guide

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1. Introduction

This guide explains how to use CSS variables for theming in the WebUp Industrial Engineering website. The website uses a modern CSS variable system that allows you to change colors, spacing, and other design elements from a central location.

2. CSS Variables Overview

All theme variables are defined in the `:root` selector at the top of `css/style.css`. This makes it easy to customize the entire website by changing just a few values.

3. Variable Categories

3.1. Color Palette

```

:root {
  /* Color Palette */
  --color-primary: #3d2f21;          /* Dark sepia for header/footer */
  --color-primary-light: #ddd;        /* Light color for hover states */
  --color-background: #f4f4f4;       /* Light gray background */
  --color-background-overlay: rgba(244, 244, 244, 0.85); /* Watermark overlay */
  --color-text-dark: #333;           /* Dark text color */
  --color-text-medium: #666;         /* Medium text color */
  --color-white: white;               /* Pure white */
}

```

3.2. Spacing

```

:root {
  /* Spacing */
  --spacing-xs: 15px;
  --spacing-sm: 20px;
  --spacing-md: 40px;
  --spacing-lg: 100px;
}

```

3.3. Logo Settings

```

:root {
  /* Logo */
  --logo-size: 80px;
  --logo-position-left: 20px;
}

```

3.4. Gallery Settings

```

:root {
  /* Gallery */
  --gallery-gap: 40px;
  --image-size: 250px;
  --image-padding: 15px;
  --image-rotation-small: 3deg;
  --image-rotation-large: 6deg;
}

```

3.5. Shadow Effects

```

:root {
  /* Shadows */
  --shadow-paper: 0 10px 30px rgba(0, 0, 0, 0.3), 0 1px 8px rgba(0, 0, 0, 0.2);
  --shadow-paper-hover: 0 15px 40px rgba(0, 0, 0, 0.4), 0 2px 10px rgba(0, 0, 0, 0.3);
}

```

3.6. Transitions and Typography

```

:root {
  /* Transitions */
  --transition-fast: 0.3s ease;

  /* Typography */
  --font-family: Arial, sans-serif;
  --line-height-text: 1.6;
}

```

4. Pre-Made Themes

4.1. Theme 1: Blue Industrial

Professional blue theme for tech-oriented companies.

```

:root {
  --color-primary: #1a3a52;          /* Dark blue */
  --color-primary-light: #85b3d1;    /* Light blue */
  --color-background: #f0f4f8;       /* Light blue-gray */
  --color-background-overlay: rgba(240, 244, 248, 0.85);
  --color-text-dark: #1a2b3c;
}

```

4.2. Theme 2: Modern Gray

Clean, neutral gray theme for versatile use.

```
:root {
  --color-primary: #2c3e50;      /* Dark gray-blue */
  --color-primary-light: #bdc3c7; /* Light gray */
  --color-background: #ecf0f1;   /* Very light gray */
  --color-background-overlay: rgba(236, 240, 241, 0.85);
  --color-text-dark: #2c3e50;
}
```

CSS

4.3. Theme 3: Deep Green

Natural green theme for sustainable/eco companies.

```
:root {
  --color-primary: #1e3a1a;      /* Forest green */
  --color-primary-light: #a8d5ba; /* Mint green */
  --color-background: #f1f8f4;   /* Very light green */
  --color-background-overlay: rgba(241, 248, 244, 0.85);
  --color-text-dark: #1e3a1a;
}
```

CSS

4.4. Theme 4: Warm Orange

Bold orange theme for energy and construction.

```
:root {
  --color-primary: #d35400;      /* Dark orange */
  --color-primary-light: #f39c12; /* Light orange */
  --color-background: #fef5e7;   /* Cream */
  --color-background-overlay: rgba(254, 245, 231, 0.85);
  --color-text-dark: #7d3c00;
}
```

CSS

4.5. Theme 5: Professional Navy

Corporate navy blue for finance/consulting.

```
:root {
  --color-primary: #001f3f;      /* Navy blue */
  --color-primary-light: #7fdbff; /* Sky blue */
  --color-background: #f4f7f9;   /* Light blue-white */
  --color-background-overlay: rgba(244, 247, 249, 0.85);
  --color-text-dark: #001f3f;
}
```

CSS

5. Applying a Theme

5.1. Method 1: Direct Replacement

1. Open `css/style.css`
2. Find the `:root` section at the top of the file
3. Replace the color variables with your chosen theme values
4. Save and refresh your browser

5.2. Method 2: Separate Theme File

Create a separate CSS file for easy theme switching:

1. Create `css/theme-blue.css`:

```
/* Blue Industrial Theme Override */
:root {
  --color-primary: #1a3a52;
  --color-primary-light: #85b3d1;
  --color-background: #f0f4f8;
  --color-background-overlay: rgba(240, 244, 248, 0.85);
}
```

CSS

2. Link it in your HTML after the main stylesheet:

```
<link rel="stylesheet" href="css/style.css">
<link rel="stylesheet" href="css/theme-blue.css">
```

HTML

5.3. Method 3: Dark Mode Support

Add automatic dark mode detection:

```
/* Light mode (default) */
:root {
  --color-primary: #3d2f21;
  --color-background: #f4f4f4;
  --color-text-dark: #333;
}

/* Dark mode */
@media (prefers-color-scheme: dark) {
  :root {
    --color-primary: #1a1a1a;
    --color-background: #2a2a2a;
    --color-text-dark: #ffffff;
    --color-text-medium: #cccccc;
    --color-background-overlay: rgba(42, 42, 42, 0.85);
  }
}
```

CSS

6. Customization Examples

6.1. Adjusting Spacing

Make the layout more compact:

```
:root {
  --spacing-xs: 10px;      /* Reduced from 15px */
  --spacing-sm: 15px;      /* Reduced from 20px */
  --spacing-md: 30px;      /* Reduced from 40px */
  --spacing-lg: 80px;      /* Reduced from 100px */
}
```

CSS

6.2. Changing Logo Size

Increase logo prominence:

```
:root {
  --logo-size: 120px;      /* Increased from 80px */
  --logo-position-left: 30px; /* Adjust position if needed */
}
```

CSS

6.3. Modifying Gallery Layout

Create a tighter gallery with larger images:

```
:root {
  --gallery-gap: 30px;      /* Reduced from 40px */
  --image-size: 300px;      /* Increased from 250px */
  --image-padding: 20px;    /* Increased from 15px */
}
```

CSS

6.4. Adjusting Image Rotation

More dramatic paper effect:

```
:root {
  --image-rotation-small: 5deg; /* Increased from 3deg */
  --image-rotation-large: 10deg; /* Increased from 6deg */
}
```

CSS

6.5. Changing Typography

Use a different font and spacing:

```

:root {
  --font-family: 'Helvetica Neue', Helvetica, Arial, sans-serif;
  --line-height-text: 1.8;      /* Increased from 1.6 */
}

```

CSS

7. Accessibility Considerations

When creating custom themes, ensure:

1. **Color Contrast:** Maintain WCAG AA minimum contrast ratio of 4.5:1
2. **Test Tools:** Use <https://webaim.org/resources/contrastchecker/>
3. **Text Readability:** Ensure text is readable on all backgrounds
4. **Focus States:** Verify link hover colors are visible

7.1. Checking Contrast

Use this formula for your colors:

```

Text on Background:
- Normal text: 4.5:1 minimum
- Large text (18pt+): 3:1 minimum

```

TEXT

Example contrast check:

```

/* Good contrast */
--color-text-dark: #333;      /* On #f4f4f4 background = 9.7:1 ✓ */
--color-text-medium: #666;    /* On #f4f4f4 background = 5.7:1 ✓ */

/* Poor contrast - avoid */
--color-text-light: #ddd;     /* On #f4f4f4 background = 1.2:1 ✗ */

```

CSS

8. Tips and Best Practices

1. **Document Changes:** Keep notes of custom theme values
2. **Test Across Browsers:** Verify appearance in Chrome, Firefox, Safari, Edge
3. **Use Hex Colors:** More precise than named colors
4. **Group Related Variables:** Keep color variables together
5. **Maintain Consistency:** Use variables instead of hardcoded values
6. **Version Control:** Commit theme changes to Git
7. **Create Theme Library:** Save successful themes for reuse

9. Responsive Theme Adjustments

Adjust variables for different screen sizes:

```

/* Mobile adjustments */
@media (max-width: 768px) {
  :root {
    --spacing-lg: 60px;      /* Reduce left margin */
    --logo-size: 60px;       /* Smaller logo */
    --image-size: 200px;     /* Smaller gallery images */
    --gallery-gap: 20px;     /* Tighter spacing */
  }
}

/* Large screens */
@media (min-width: 1920px) {
  :root {
    --spacing-lg: 150px;     /* More breathing room */
    --image-size: 300px;     /* Larger gallery images */
    --gallery-gap: 60px;     /* More space */
  }
}

```

CSS

10. Troubleshooting

10.1. Theme Not Applying

Problem: Changes to variables don't appear

Solutions:

1. Clear browser cache (Ctrl+Shift+R or Cmd+Shift+R)
2. Check CSS file is linked correctly in HTML
3. Verify no syntax errors in CSS (missing semicolons, etc.)
4. Ensure `:root` selector is at the top of the file

10.2. Colors Look Wrong

Problem: Colors appear different than expected

Solutions:

1. Check monitor color calibration
2. Test in multiple browsers
3. Verify rgba alpha values (0.85 = 85% opacity)
4. Check for CSS specificity issues

10.3. Variables Not Supported

Problem: Old browser doesn't support CSS variables

Solutions:

1. Add fallback values:

```
.content {  
  background-color: #f4f4f4;           /* Fallback */  
  background-color: var(--color-background); /* Modern browsers */  
}
```

2. Target browsers: Chrome 49+, Firefox 31+, Safari 9.1+, Edge 15+

11. Additional Resources

- MDN CSS Variables: https://developer.mozilla.org/en-US/docs/Web/CSS/Using_CSS_custom_properties
- Color Contrast Checker: <https://webaim.org/resources/contrastchecker/>
- Color Palette Generator: <https://coolors.co/>
- WCAG Guidelines: <https://www.w3.org/WAI/WCAG21/quickref/>

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